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Integration of photovoltaic systems in Algeria's building electrical network: technical and financial analysis

Integração de sistemas fotovoltaicos na rede elétrica de edifícios da Argélia: análise técnico-econômica

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ABSTRACT
This extensive study offers a thorough analysis of the complex technical and economical factors involved in incorporating photovoltaic (PV) systems into the buildings' electrical grid in Algeria. It focuses on a crucial element in the country's energy transition. This study aims to evaluate the viability and potential advantages of implementing solar panels on the rooftop of a university parking lot located at the prestigious Faculty of Electrical Engineering of the University of Sidi Bel Abbess. By utilising advanced modelling tools such as PVSOL and HOMER Pro, this study applies a comprehensive methodology to assess the feasibility of this facility. In the first step, a comprehensive three-dimensional model is carefully constructed using PVSOL software in order to assess the maximum achievable photovoltaic (PV) capacity for the neighbouring parking lot. This estimation considers several parameters, including potential shadows, size of furniture, direction, and slope. This thorough assessment establishes the foundation for subsequent examinations. Following this, a thorough technical-economic analysis is performed using HOMER Pro software to comprehensively evaluate the advantages and disadvantages of two separate scenarios: one that exclusively depends on the current electrical network and another that combines conventional grid power with solar-generated electricity. The study offers useful insights into the economic feasibility and sustainability of incorporating photovoltaic (PV) systems into the university's infrastructure, taking into account several aspects such as installation costs, maintenance requirements, electricity rates, and possible savings. The results highlight the considerable potential offered by integrating photovoltaic (PV) systems into the electrical grids of buildings, enabling the provision of electricity to both structures and perhaps facilitating the recycling of electric vehicles. Not only does this have the potential to reduce long-term energy costs, but it also plays a crucial role in promoting Algeria's shift towards more eco-friendly and sustainable energy options. This, in turn, contributes to global initiatives to address climate change and promote a more environmentally friendly future for future generations.

Keywords: photovoltaic systems, grid integration, technical-economic analysis, 3D simulation, Algeria.

RESUMO
Este extenso estudo oferece uma análise detalhada dos complexos fatores técnicos e econômicos envolvidos na incorporação de sistemas fotovoltaicos (PV) na rede elétrica dos edifícios na Argélia. Ele foca em um elemento crucial na transição energética do país. Este estudo tem como objetivo avaliar a viabilidade e os potenciais benefícios da implementação de painéis solares no telhado de um estacionamento universitário localizado na prestigiada Faculdade de Engenharia Elétrica da Universidade de Sidi Bel Abbess. Utilizando ferramentas de modelagem avançadas como PVSOL e HOMER Pro, este estudo aplica uma metodologia

abrangente para avaliar a viabilidade desta instalação. Na primeira etapa, um modelo tridimensional abrangente é cuidadosamente construído usando o software PVSOL para avaliar a capacidade fotovoltaica (PV) máxima alcançável para o estacionamento vizinho. Esta estimativa considera vários parâmetros, incluindo sombras potenciais, tamanho dos móveis, direção e inclinação. Esta avaliação minuciosa estabelece a base para exames subsequentes. Seguindo isso, uma análise técnico-econômica completa é realizada usando o software HOMER Pro para avaliar de forma abrangente as vantagens e desvantagens de dois cenários separados: um que depende exclusivamente da rede elétrica atual e outro que combina energia elétrica convencional da rede com eletricidade gerada por energia solar. O estudo oferece insights úteis sobre a viabilidade econômica e sustentabilidade da incorporação de sistemas fotovoltaicos (PV) na infraestrutura da universidade, levando em consideração vários aspectos como custos de instalação, requisitos de manutenção, tarifas de eletricidade e possíveis economias. Os resultados destacam o considerável potencial oferecido pela integração de sistemas fotovoltaicos (PV) nas redes elétricas dos edifícios, permitindo o fornecimento de eletricidade para ambas as estruturas e talvez facilitando o recarregamento de veículos elétricos. Isso não apenas tem o potencial de reduzir os custos de energia a longo prazo, mas também desempenha um papel crucial na promoção da mudança da Argélia para opções de energia mais ecológicas e sustentáveis. Isso, por sua vez, contribui para iniciativas globais de combate às mudanças climáticas e promoção de um futuro mais ambientalmente amigável para as gerações futuras.

Palavras-chave: sistemas fotovoltaicos, integração à rede, análise técnico-econômica, simulação 3D, Argélia.

1 INTRODUCTION

Globally, there is a growing trend toward renewable energy sources as a means of mitigating climate change and reducing greenhouse gas emissions (Kazem, 2011).The International Energy Agency projects that photovoltaics (PV) will surpass all other power sources worldwide by the middle of the century (CHEN *et al.,* 2021).Cost savings, technological developments, and pro-PV legislation are driving growth in PV; in many places, PV is already in competition with fossil fuels (Rahman et al., 2022)

Grid-connected, hybrid, and freestanding photovoltaic systems will all be essential to the transition to a sustainable environment in the future (Obaideen *et al.,* 2023).Communities, small businesses, and end users of power can generate energy from solar radiation with grid-connected photovoltaic systems. Surplus energy is replenished back into the grid. This strategy can reduce energy expenses and dependence on fossil fuels (Mukisa *et al.,* 2022). In the event of a grid outage, grid-connected PV systems may also serve as a reliable power source, especially if storage devices are installed to provide the necessary electricity.

Like many other developing nations, the government of Algeria has established the "National Program for Renewable Energies and Energy Efficiency" as an initial step towards sustainability. As per national policy, by 2030, 40% of Algeria's electricity consumption is expected to be sourced from renewable energy (Ghezloun *et al.,* 2015). Due to its abundant solar radiation, covering 2,382 million km^2 and 90% of the nation, Algeria has pledged to primarily rely on solar photovoltaics (PV) for its renewable energy needs (Razagui *et al.,* 2017). According to reference (Abdeladim *et al.,* 2014), there are approximately 3000 hours of sunshine per year, with daily energy levels reaching up to 5 kWh/m^2.

Within the context of this energy transition, research and concrete initiatives will be necessary to achieve these objectives. To assess Algeria's potential for renewable energy and integrate solar systems into the network, several studies have been conducted. Studies such as those by (Attari *et al.,* 2016; Bouacha *et al.,* 2020),for example, have examined how Algeria's solar potential and photovoltaic performance are evaluated. The foundation for comprehending the advantages and challenges associated with the country's integration of renewable energy sources has been established by this research.

The world to evaluate the possibilities of renewable energy sources and create networks with integrated photovoltaic systems. For example, studies conducted by academics in the United States (Imam *et al.,* 2019) have examined the effectiveness of home solar systems in various temperature zones. Furthermore, research carried out in China (Peidong *et al.,* 2009)has explored the advantages and disadvantages of integrating solar energy into metropolitan electrical networks.

However, determining the size and technical and economic viability of a photovoltaic system connected to the network presents a challenging task involving a number of factors, including site isolation, the best panel orientation and tilt, the effects of shading from the environment, building energy consumption, and the costs associated with installation, operation, and maintenance (Ramadhan; Naseeb, 2011)Consequently, to ensure the feasibility and profitability of such a project, a rigorous methodology based on advanced modeling tools is required (Gutiérrez Galeano *et al.,* 2018)

The primary aim of this research is to determine the optimal dimensions of a grid-connected photovoltaic system capable of supplying electricity to three distinct loads: the potential for vehicle charging and two university facilities. This study employs the PVSOL program to generate a three-dimensional model for estimating the optimal quantity of solar panels that can be affixed to the roof of the adjacent parking lot. Subsequently, employing the HOMER Pro software, a comprehensive evaluation will be conducted to compare the technical and economic aspects of a system that solely relies on the building's electrical network and a system that integrates both the electrical network sources and the solar panels affixed to its roof.

2 GEOGRAPHICAL CHART OF THE STUDY SITE.

Implantation Site: The selected site is an unutilized parking lot located next to two university buildings at the Faculty of Electrical Engineering at the University of Sidi Bel Abbès (Figure 1). The geographic coordinates of this place are 35°12'59.9"N latitude and 0°38'40.7"W longitude.

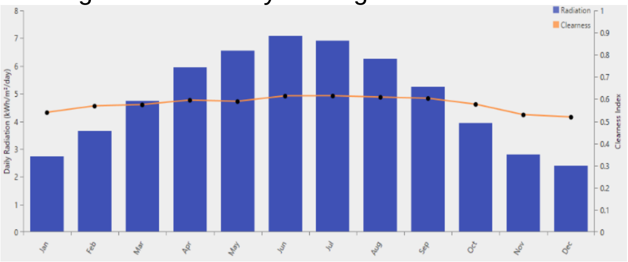
Figure 1 – Location of Implantation

Source: Authors

Solar radiation: Figure 2 on our website displays the monthly data for solar radiation and the clarity index for the full year. The climatically varied areas exhibit seasonal patterns that result in solar radiation values reaching their highest point during the summer months (May to August, with a maximum in July) and reaching their lowest levels during winter (November to February, with a minimum in December). The transparency index, a metric used to evaluate the cleanliness of the atmosphere and its capacity to reflect solar radiation, has a similar pattern, measuring 0.4% during winter and 0.9% during summer.

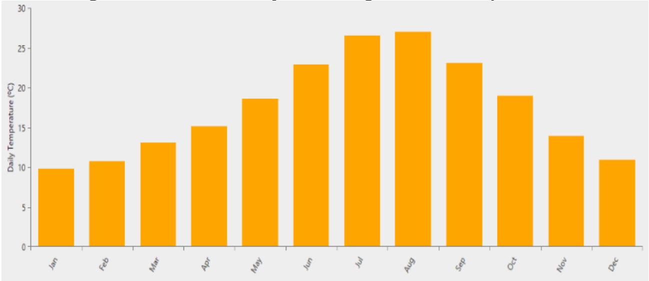
Temperature: The average monthly temperature data from our geography website is shown in Figure 3, whereby the orange bars correspond to degrees Celsius (°C). The period spanning from June to August is characterized by the highest temperatures, with an average maximum of 25°C recorded in July. Conversely, January often experiences temperatures as low as around 5°C during the winter season, with December to February being the coldest months. The temperature trend seen in several temperate nations aligns with the typical warm summers and cold winters. The transitional months of April, May, September, and October often exhibit average temperatures ranging from 10°C to 15°C on the Celsius scale.

Figure 2 – Monthly Average Solar GHI Data



Source: Authors

Figure 3 – Monthly Average air Temperature

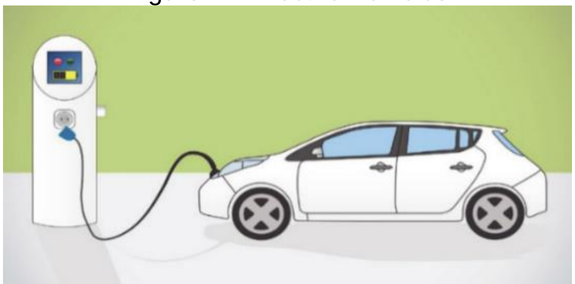


Source: Authors

Characteristics of the chosen charge: The scope of our proposal includes the potential inclusion of charging stations for select automobiles as well as the development of two academic infrastructures. Electric vehicle charging stations: The parking roof is equipped with a solar field, enabling the potential for vehicle recharging (Figure 4). Electric vehicles are few in Algeria, particularly in the city of

Sidi Bel Abbess. Consequently, a reduction in the number of vehicles requiring recharging will lead to a corresponding decrease in energy usage (Mougay, 2021).

Figure 4 – Electric Vehicles



Source: MOUGAY, 2021

The first building: The first structure accommodates a library equipped with a total of 270 lights, 145 computers, and a limited number of monitoring equipment. The library is often used for the purpose of completing practice assignments and attending conferences, leading to a total of 8 hours of morning usage and 17 hours of evening utilization. In contrast, the morning is often characterized by a reduced number of activities, leading to decreased energy expenditure during that specific time frame (Figure 5).

Figure 5 – First building (library)

Source: Authors

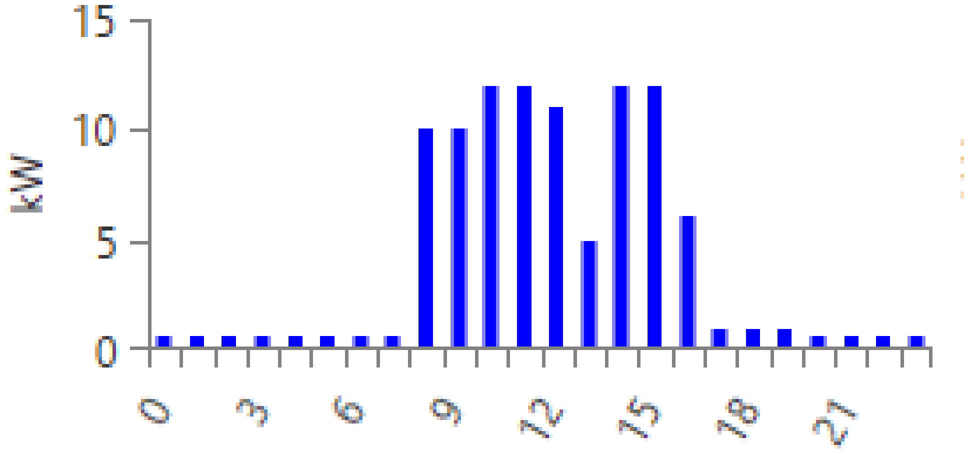
The second building: The second edifice functions as an administrative edifice with many levels, each housing office spaces and conference rooms furnished with essential amenities. Although these two infrastructures are crucial elements of the institution, they are often not used on weekends or holidays, save for a few lights for security officers (Figure 6).

Load Profile (Daily): The energy consumption profile for the three loads during a 24-hour period is shown in Figure 7. Two clearly distinguishable peaks can be observed: the first peak, which takes place at roughly 11 a.m., exhibits a power output of approximately 13 kW as a result of heightened daytime activity. Similarly, the second peak, which emerges around 3 p.m., similarly demonstrates a power output of approximately 13 kW owing to afternoon activities, suggesting uninterrupted use throughout office hours. Consumption exhibits a decline at around 1 p.m. as a result of the breakfast break. The demand decreases to around 1 kilowatt during the nighttime hours, namely from 11 p.m. to 6 a.m., which corresponds to closure times.

Figure 6 – Second building (Tower)

Source: Authors

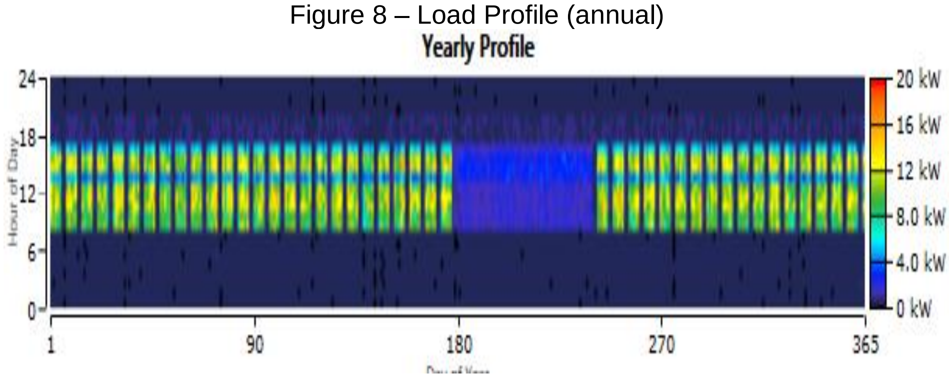
Figure 7 – Load Profile (Daily)



Source: Authors

Load Profile (annual): The monthly power use of the two buildings during a whole year is shown in Figure 8. A significant decrease in consumption is seen throughout the months of July and August, as well as in December. These declines may be attributed to the university vacation times, during which there is less activity in buildings. In contrast, throughout the remaining months of the academic year, there is a notable increase in consumption. This may be attributed to the normal functioning of educational institutions, which include the presence of staff, students, and many activities that need the use of electricity.

Figure 8 – Load Profile (annual)



Source: Authors

3 ANALYSIS TOOLS

PVSOL software: PVSOL, a solar modeling program, was created by Valentin program GmbH, a German company. The design, conception, and modeling of photovoltaic (PV) solar energy systems are often conducted within the solar energy industry. By using the advanced functionalities and intuitive interface of PVSOL, professionals are able to evaluate the solar capacity of a given location, quantify the efficiency of photovoltaic (PV) systems, simulate their operational effectiveness, and compute their financial profitability (Ibrahim, 2022; Manikandan *et al.,* 2023).

HOMER software: HOMER Pro is a very advanced modeling and optimization program designed for the investigation of decentralized and hybrid energy systems. This tool enables users to assess the technical and economic viability of energy system designs, taking into account a range of energy sources including diesel generators, solar panels, wind turbines, and hydroelectric power. The program employs advanced optimization algorithms to determine the most economically efficient and lucrative solutions, taking into account the user's unique objectives, such as reducing costs, enhancing energy supply dependability, or mitigating greenhouse gas emissions (Alhousni *et al.,* 2023).

The modeling of a photovoltaic cell: The following equation is used to determine the photovoltaic matrix's output power (Maoulida *et al.,* 2022).

$$P_{PV} = Y_{PV} f_{PV} \left(\frac{G_T}{G_{T.STC}}\right) [1 + \propto_P (T_C - T_{C.STC})] \quad (1)$$

Y_{PV} Nominal PV generator capacity [kW].
f_{PV} Declassification factor PV, [%].
$\propto_P$ Temperature coefficient of the power [%/°C].
T_C Temperature of the PV cell, [°C].
$T_{C.STC}$ PV temperature under normal conditions: [25°C]

If an individual opts not to incorporate the influence of temperature on the photovoltaic generator, the aforementioned equation (1) becomes more straightforward.

$$P_{PV} = Y_{PV} f_{PV} \left(\frac{G_T}{G_{T.STC}}\right) \quad (2)$$

4 RESULTATS AND DISCUSSIONS

Simulation with Pvsol: Figure 9 depicts a three-dimensional representation of the photovoltaic system situated on the parking lot beneath the covered overhanging parking lots. The solar panels exhibit a distinct visual appeal due to their vertical arrangement, enabling the placement of 408 panels with three panels per vertical section and appropriate spacing between panels over different sections of the roof. The utilization of 3D modeling proven to be highly advantageous in the assessment of the solar capacity of a given location and the identification of the optimal dimensions for the photovoltaic system. It allows for considering crucial factors such as tree orientation, potential air barriers such as vegetation or other buildings, and the optimal angle for solar panels to maximize energy production.

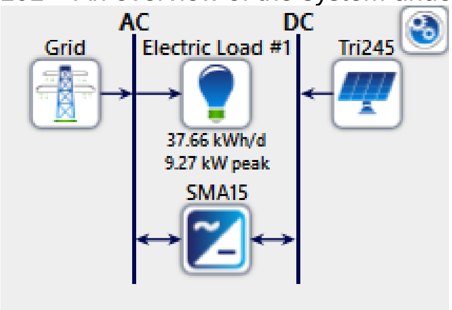
Figure 9 – Parking lot's 3D design

Source: Authors

4.1 SIMULATION WITH HOMER

An overview of the system under study: Using the PVsol software, we were able to build 408 solar panels while accounting for lighting and shading, allowing us to analyze the technical and financial elements of this system. The system under study consists of six SMA15 DC-AC converters, each costing 3566.17 euros, solar generators of the Trina Solar 245TSM type with a capacity of 245 watts each cost 75 euros, a public network, and electricity costs related to the equipment and operating hours, as shown in Figure 10.

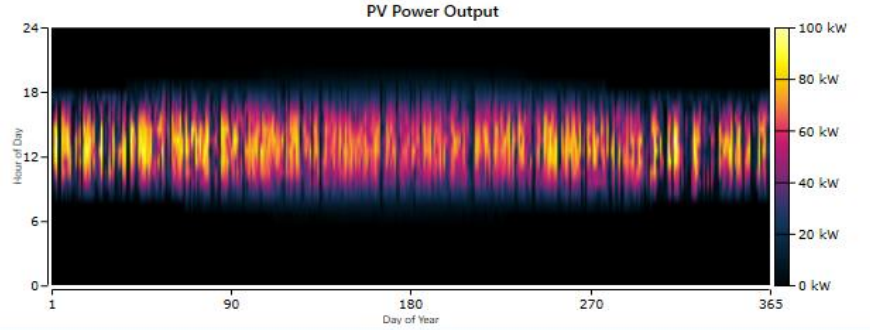
Figure 101 – An overview of the system under study



Source: Authors

Pv Power Output: The electricity output profile of a solar system during a 24-hour period is depicted in Figure 11. Pursuant to the solar cycle, production ceases during nighttime, surges in the early morning as the sun ascends, attains a maximum of approximately 90 kW by midday when solar radiation is at its zenith, subsequently diminishes gradually towards the conclusion of the afternoon, and ultimately reverts to zero during solar radiation. The sporadic production pattern is logical considering the consistent presence of solar energy during a normal day.

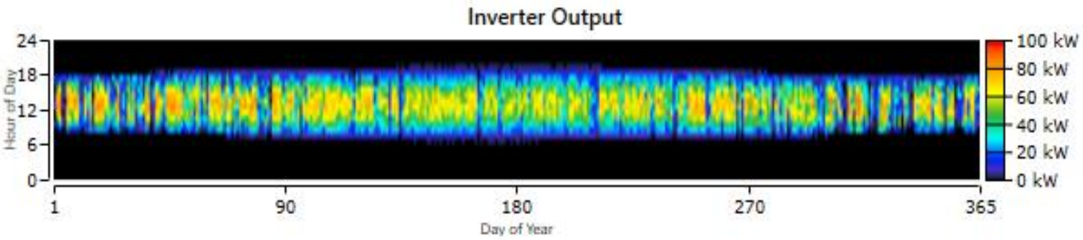
Figure 11 – Pv Power Output



Source: Authors

Inverter output: Figure 12 illustrates the power output of the converters within the photovoltaic system during a 24-hour duration. A curve resembling the photovoltaique production can be observed, as depicted in Figure 3. This curve exhibits zero power during the night, followed by a gradual increase in the morning, reaching its highest point around midday, which corresponds to the maximum solar radiation. Subsequently, the power decreases in the afternoon until it reaches zero once again at night. The depicted curve accurately represents the electrical output produced by solar panels, which has been subsequently transformed using CC/CA converters or microgrids.

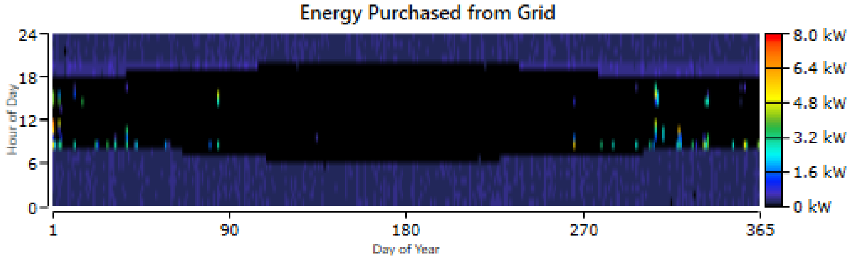
Figure 12 – Inverter output



Source: Authors

Energy purchased from grid: In Figure 13, the data illustrates the quantity of electricity acquired by the distribution network over a span of 24 hours. During the nocturnal period, specifically from approximately 18 to 8 AM, when the photovoltaic panels are not producing any electricity, it is possible to observe the acquisition of photographs. These graphics depict situations where, when there is no local solar power available, the building's electricity has to be completely satisfied by buying energy from the power grid. The insufficient levels of consumption in photovoltaic generation can be attributed to the relatively lower purchasing points observed during midday.

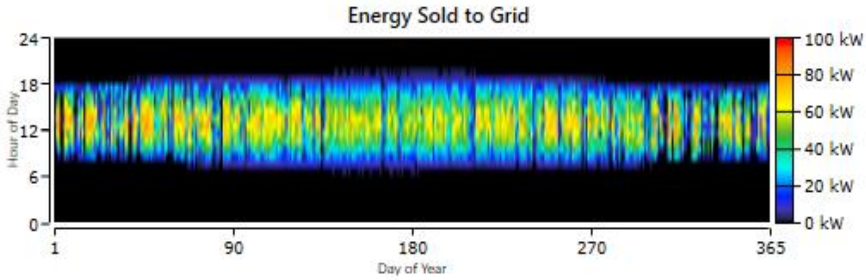
Figure 13 – Energy purchased from grid



Source: Authors

Energy sold to grid: Figure 14 illustrates the distribution network's electrical energy sales profile over a 24-hour duration. During the period of darkness, typically spanning from 8 a.m. to 18 p.m., visual representations of energy sales can be observed. The aforementioned photographs pertain to instances wherein the quantity of electricity generated by solar panels surpasses the quantity of electricity demanded by structures. then, the surplus production is introduced and then traded inside the network. Upon observation, it is evident that there is a lack of sales during the nighttime period when the solar panels are not generating any energy. The graph illustrates instances in which the building's energy usage is below the level of photovoltaic generating.

Figure 14 – Energy sold to grid

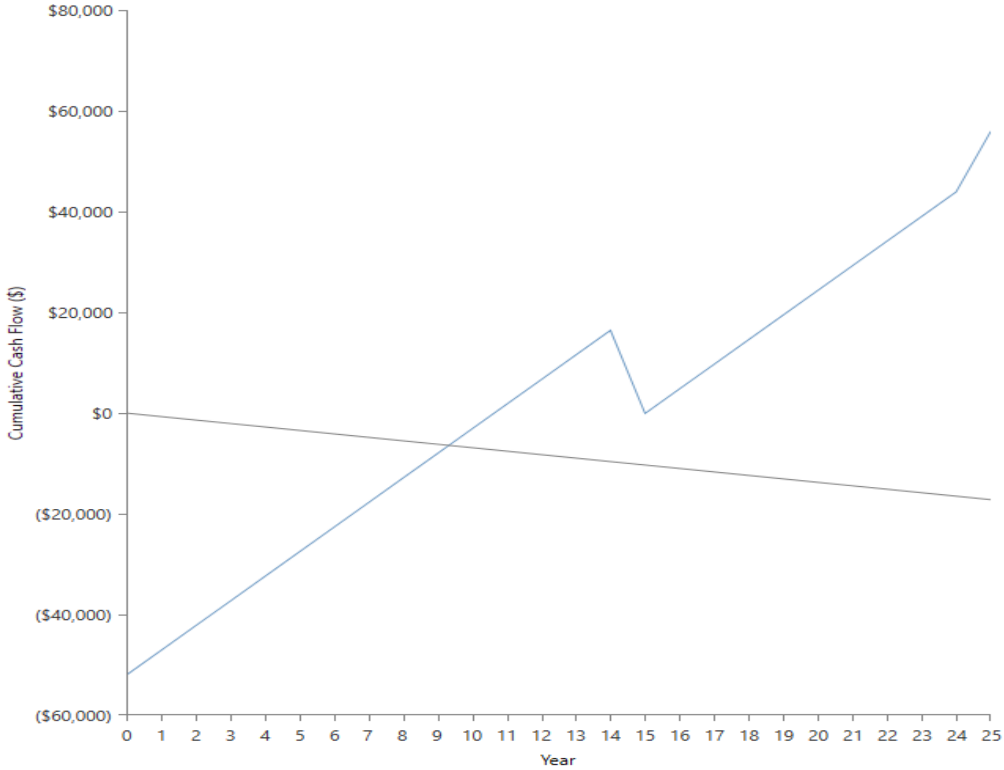


Source: Authors

Techno-economic comparison of the two system: Figure 15 illustrates a comparative analysis of the cumulative costs incurred over a span of 25 years for two distinct scenarios pertaining to the electrical supply for our load. The gray curve depicts the aggregate expenses incurred when the load is only provided with power through the grid. The cost representation in the blue curve pertains to the integration of a solar system with the grid in a hybrid system. The hybrid system necessitates a considerably greater initial capital outlay, specifically around 50,000 euros, accompanied by elevated expenses over the initial years. Nevertheless, after around a decade, the cost curve of the hybrid system coincides with that of the grid alone, resulting in increased profitability for the hybrid system. As a result, a subsequent investment is made after a period of 15 years, leading to a decline in the curve as a result of reinvestments in inverters that have a lifespan of 15 years. Therefore, over a span of 25 years, the hybrid scenario proves to be more financially advantageous than the grid-only supply, with a significant profit margin.

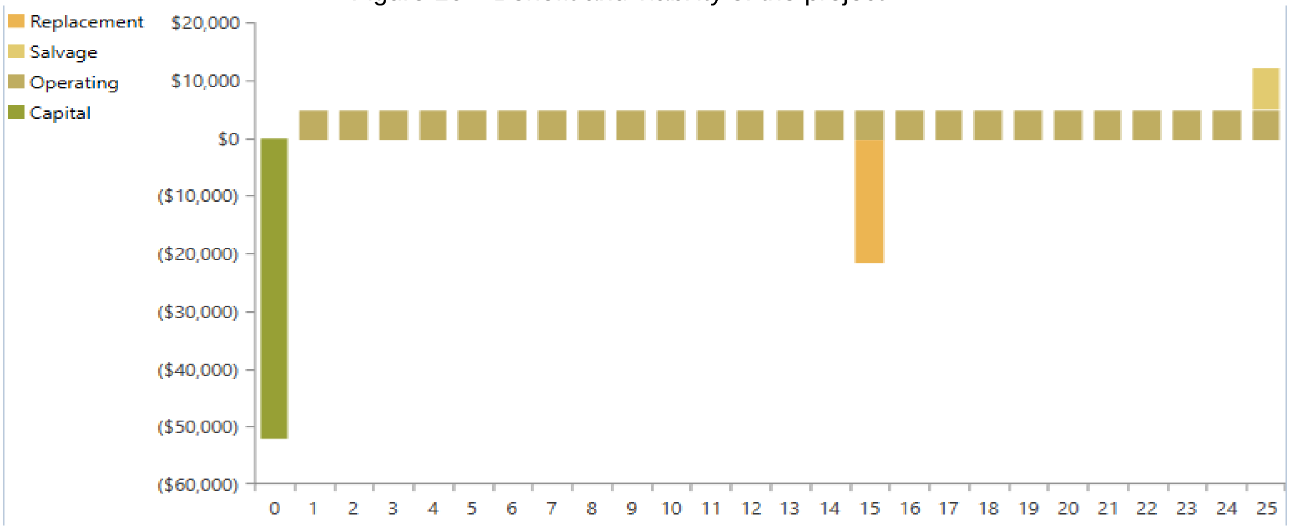
Benefit and viability of the project: Figure 16 enumerates the diverse expenses associated with the acquisition and functioning of a photovoltaic system linked to the electrical grid across a span of 25 years. The green bars displayed in the graph represent the initial capital expenditures linked to the implementation of solar panels and accompanying equipment, including converters. The orange bars depict the expenses associated with replacing components that have a restricted lifespan. It is evident that there is a single expense associated with replacing the transformer in the fifth year. The recurring little bars represent the annual operation and maintenance expenses of the system. The residual value of the equipment that remains in good condition after a period of 25 years is represented by the yellow bar located at the border of the graph.

Figure 15 – Techno-economic comparison of the two system



Source: Authors

Figure 16 – Benefit and viability of the project



Source: Authors

5 CONCULSION

Concisely, this research significantly improves our understanding of how photovoltaic (PV) systems are integrated into the electrical grid of buildings in Algeria. We have thoroughly assessed the feasibility and potential benefits of this integration by use sophisticated modeling tools such as PVSOL and HOMER Pro. The research's findings offer a comprehensive analysis of the potential impacts of incorporating photovoltaic systems into Algerian buildings on both society and academia. They exemplify the enduring economic advantages of this approach, encompassing a reduction in energy expenditures and a transition towards sustainable and renewable energy sources. This transition reduces Algeria's dependence on imported hydrocarbon fuels and contributes to the global effort to mitigate climate change, addressing both the present and future energy requirements of the country. Within the academic context, the utilization of these systems can serve as a prototype and educational tool for scholars and learners investigating sustainable development and renewable energy. Nevertheless, it is necessary to recognize various constraints of this study. The examination was primarily carried out on a specific site characterized by distinct climate and sunlight conditions. Further investigation is required to ascertain the feasibility of implementing similar projects in other regions of Algeria that possess varying factors. Furthermore, the study solely focused on the installation of photovoltaic panels on parking shade, neglecting alternative options such as solar-powered building facades or rooftops. These alternatives might potentially enhance the campus's renewable energy generation capacity. Furthermore, a comprehensive technical-economic analysis could explore more intricate situations, such as the incorporation of energy storage devices or other energy sources. Additional research in these fields is necessary to better substantiate the feasibility and appeal of integrating photovoltaic projects into university buildings in Algeria.

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